

Experiment 'p_6_2_c_replace_even_with_zero_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_6_2_c_replace_even_with_zero_seq

Natural language statement: null

Method signature: p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `forall i :: 0 <= i < |inputs| ==> (inputs[i] % 2 == 0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

ensures `|result| == |inputs|`, ensures forall `i :: 0 <= i < |inputs| ==> (inputs[i] % 2 == 0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNjZVNg5gge4bYkTfNjCDwri1CZt

Model: gpt-5.5-2026-04-23

Created at: 1780661677

Finish reason: stop

Usage: 362 tokens in, and 489 tokens out

```
{
  result := seq(|inputs|, i => if inputs[i] % 2 == 0 then 0 else inputs[i]);
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_6_2_c_replace_even_with_zero_seq/program.dfy(5,40): Error: index out of
  ↪ range
|
5 | result := seq(|inputs|, i => if inputs[i] % 2 == 0 then 0 else inputs[i]);
  | ^
```

Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the Dafny body, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNkILfRGAhjV7vo0osq0DISyITy

Model: gpt-5.5-2026-04-23

Created at: 1780661722

Finish reason: stop

Usage: 545 tokens in, and 1568 tokens out

```
{
  result := seq(|inputs|, i requires 0 <= i < |inputs| => if inputs[i] % 2 == 0 then 0
    ↪ else inputs[i]);
}
```

Final Program

```
method p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)
  ensures |result| == |inputs|
  ensures forall i :: 0 <= i < |inputs| ==> (inputs[i] % 2 == 0 ==> result[i] == 0)
    ↪ && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])
{
  result := seq(|inputs|, i requires 0 <= i < |inputs| => if inputs[i] % 2 == 0 then 0
    ↪ else inputs[i]);
}
```

Total Token Usage

Input tokens: 907

Output tokens: 2057

Reasoning tokens: 1939

Sum of ‘total tokens’: 2964

Experiment Timings

Overall Experiment started at 1780661676640, ended at 1780661780230, lasting 103590ms (103.59 seconds)

Iteration #1 started at 1780661676641, ended at 1780661721980, lasting 45339ms (45.34 seconds)

Iteration #2 started at 1780661722017, ended at 1780661780230, lasting 58213ms (58.21 seconds)